

Joint Lung + Nodule Segmentation — SegResNet (3D, ex-LIDC)

1. Task

Voxel-level joint segmentation of lung tissue and pulmonary nodules in 3D chest CT volumes. Single model, single forward pass, no ROI stage and no bounding-box crop.

- **Input** (1, 256, 256, 256) full CT resampled to 256^3 , intensity-normalised to [0, 1].
- **Output** (3, 256, 256, 256) softmax logits: class 0 = background, class 1 = lung, class 2 = nodule.

A voxel that is both lung tissue and nodule is assigned exclusively to class 2 (nodule takes precedence over lung) since the head uses a mutually-exclusive softmax.

2. Model

[MONAI SegResNet](#), SegResNet (residual U-Net).

Property	Value
Architecture family	SegResNet (residual U-Net)
Spatial dims	3
Input channels	1
Output channels	3 (softmax)
Initial feature width	16
Encoder blocks per level	[1, 2, 2, 4, 4]
Decoder blocks per level	[1, 1, 1, 1]
Dropout	0.1
Trainable parameters	20,663,555

3. Data

Trained on the `unified_v2_ex_lidc` split (patient-grouped, dataset-stratified, **LIDC excluded**):

Source	Role
NLST	train + val + test
NSCLC-Radiomics	train + val + test

Source	Role
~~LIDC-IDRI~~	<i>excluded</i>

Split sizes: **1 110 train / 196 val / 129 test (held out)**.

Lung labels are ground-truth per-slice 2D masks (`roi_seg_seg_2d/`), stacked into 3D per series. Nodule labels come from the corpus's own 3D nodule annotations.

4. Training

Setting	Value
Loss	Multi-class Focal Tversky + weighted CE ($\alpha=0.3$, $\beta=0.7$, $\gamma=2.0$, $\lambda_{CE}=0.1$, class weights = <code>[1.0, 1.0, 100.0]</code> for <code>[bg, lung, nodule]</code>)
Optimizer	Adam
Learning rate	1×10^{-5}
Weight decay	1×10^{-5}
LR schedule	Cosine annealing, $T_{max} = 400$, $\eta_{min} = 1 \times 10^{-6}$
Batch size	4
Epochs	400
Random seed	42
Mixed precision	bf16 (autocast, no GradScaler)
Augmentation	3D flips, 90° rotations, elastic rotation, zoom, intensity scale/shift, Gaussian noise/blur, contrast
Hardware	1 × NVIDIA H100 94 GB
Wall-clock	≈ 3.5 days

Best checkpoint (evaluated below): **epoch 209 / 400**.

5. Evaluation protocol

Evaluated on the val split (196 series, never seen during training). Predictions taken as `argmax` over the 3-channel softmax output. All per-voxel metrics are micro-averaged over the whole split (3,288,334,336 voxels total).

6. Results

Ticket-required metrics

Metric	Value
mean IoU (all 3 classes)	0.8147
mean IoU (lung + nodule)	0.7250
Accuracy	0.9947
Precision (macro, all classes)	0.8766
Precision (macro, non-bg)	0.8157
Recall (macro, all classes)	0.8788
Recall (macro, non-bg)	0.8204

Per-class breakdown

Class	IoU	Precision	Recall	Dice (micro)
Background	0.9940	0.9985	0.9955	0.9970
Lung	0.9635	0.9725	0.9905	0.9814
Nodule	0.4865	0.6589	0.6503	0.6546

Per-case Dice distribution (per class, 196 val cases)

Class	Mean	Std	p25	Median	p75
Background	0.9970	0.0028	0.9968	0.9971	0.9975
Lung	0.9775	0.0199	0.9762	0.9835	0.9857
Nodule	0.5324	0.2491	0.3815	0.5803	0.7291

7. Notes on interpretation

The joint task's class distribution is heavily skewed: background dominates, lung tissue is ~14 % of the volume, and nodule voxels are on the order of 10^{-5} . This shapes what each metric means here:

- **Accuracy** looks near-perfect (~ 0.99) because getting background right accounts for most voxels regardless of model quality. Read it as a floor, not a discriminator.
- **mIoU** across all three classes is dominated by the near-1.0 `IoU_background`; the *no-bg* variant (`IoU_lung`, `IoU_nodule` averaged) is the more diagnostic number for comparing the two model families.
- **Precision / Recall** for the lung head sit at ~0.98 — the model has learned lung anatomy well. The nodule head has Precision and Recall in the 0.62–0.75 range and is where architecture / data choices actually differentiate.

- **Per-case Dice** for the nodule class carries a wide distribution ($\text{std} \approx 0.25$) driven by nodule size and location — the median is a better summary than the mean for clinical use.

8. How to load & run inference

The pre-trained weights are hosted at [Kakimaki00/joint-segresnet-3d-ex-lidc](#).
Download and load directly:

```
huggingface-cli download Kakimaki00/joint-segresnet-3d-ex-lidc --local-dir ./ckpt
```

```
import yaml, torch
from monai.networks.nets import SegResNet

cfg = yaml.safe_load(open("config.yaml"))["model"]
model = SegResNet(
    spatial_dims = cfg["spatial_dims"],
    in_channels   = cfg["in_channels"],
    out_channels  = cfg["out_channels"],    # 3
    init_filters  = cfg["init_filters"],
    blocks_down   = tuple(cfg["blocks_down"]),
    blocks_up     = tuple(cfg["blocks_up"]),
    dropout_prob  = cfg["dropout_prob"],
)
state = torch.load("model.pth", map_location="cpu", weights_only=True)
model.load_state_dict(state)
model.eval()

with torch.no_grad():
    x = torch.randn(1, 1, 256, 256, 256)           # full CT resampled to 2563
    logits = model(x)                               # (B, 3, D, H, W)
    pred_class = logits.argmax(dim=1)               # (B, D, H, W) in {0, 1, 2}
    lung_mask  = (pred_class == 1).to(torch.uint8)
    nodule_mask = (pred_class == 2).to(torch.uint8)
```

Unlike the two-stage `nodule-*` HF checkpoints, this model does **not** need a lung-bbox crop — feed it the whole CT resampled to 256³.

9. Reproducibility

- Config `configs/joint_segresnet_ex_lidc.yaml`
- Checkpoint `stage3_joint/checkpoints_segresnet_ex_lidc/best_model.pth`
- Metrics JSON `stage3_joint/checkpoints_segresnet_ex_lidc/eval_metrics_joint.json`
- Command `python stage3_joint/eval_metrics.py --config configs/joint_segresnet_ex_lidc.yaml --checkpoint stage3_joint/checkpoints_segresnet_ex_lidc/best_model.pth`

Companion variant: [Kakimaki00/joint-segresnet-3d-pseudo-lidc](#) — same architecture, pseudo-LIDC data variant.